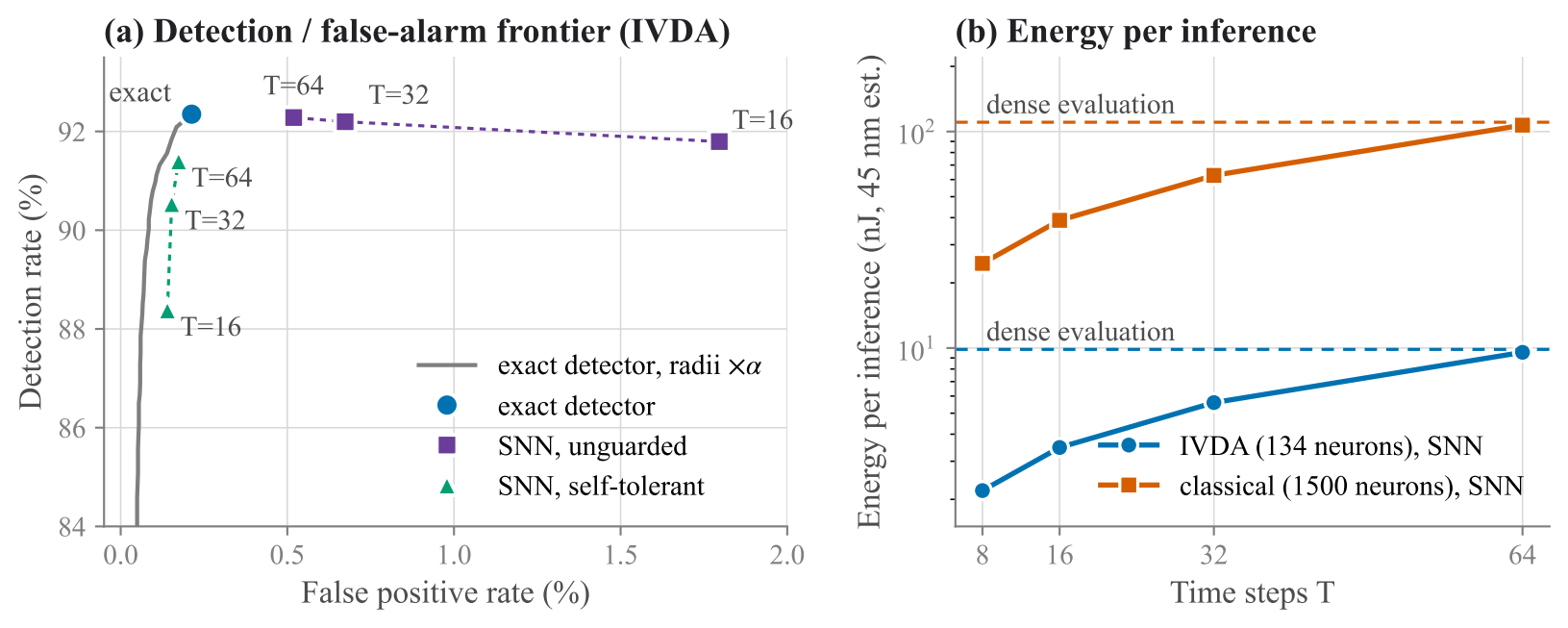
(a) Detection / false-alarm frontier (IVDA)
exact
T=64
T=32
T=16
T=64
T=32
T=16
92
90
88
86
84
Detection rate (%)
0.0
0.5
1.0
1.5
2.0
False positive rate (%)
exact detector, radii ×α
exact detector
SNN, unguarded
SNN, self-tolerant
(b) Energy per inference
dense evaluation
dense evaluation
10^2
10^1
Energy per inference (nJ, 45 nm est.)
IVDA (134 neurons), SNN
classical (1500 neurons), SNN
8
16
32
64
Time steps T